Manufacturing-Aware 2D Nesting System

Sheet Cutting Optimisation

Preliminary Project Report

Submitted by

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1. Introduction

In modern sheet-based manufacturing processes such as laser cutting, plasma cutting, and waterjet machining, efficient utilisation of raw material plays a critical role in determining production cost. The arrangement of multiple parts on a sheet, commonly referred to as nesting, is therefore an important optimisation problem.

Traditional nesting approaches primarily treat the problem as a geometric packing task, where the objective is to maximise the number of parts within a given area while avoiding overlap. While such approaches achieve high packing density, they often fail to consider key manufacturing aspects such as cutting time, toolpath efficiency, and process constraints.

This project aims to extend the conventional understanding of nesting by incorporating manufacturing considerations directly into the placement process.

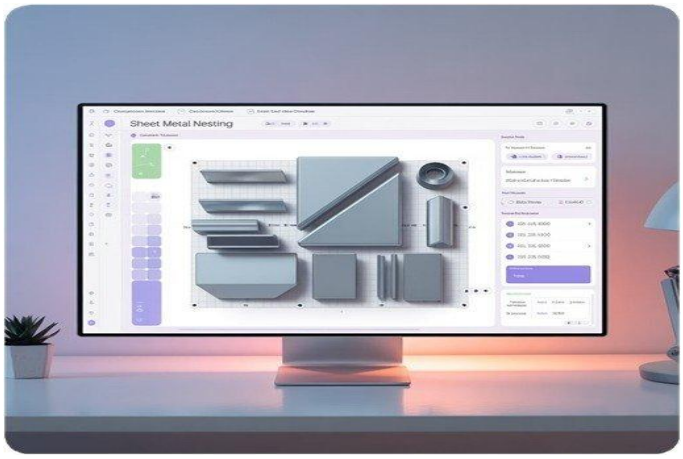


Fig: Illustration of nesting techniques

2. Problem Motivation and Gap

Most existing nesting methods optimise layouts based solely on spatial efficiency. However, real-world manufacturing introduces additional factors that significantly influence overall performance:

- Cutting time depends on total toolpath length
- Machine cost depends on operational time and number of piercings
- Constraints such as kerf width and minimum feature size affect feasibility
- Tool movement between cuts contributes to inefficiency

As a result, layouts that are geometrically optimal may not be optimal from a manufacturing perspective.

This leads to the central observation:

> *Geometric optimality does not necessarily imply manufacturing optimality.*

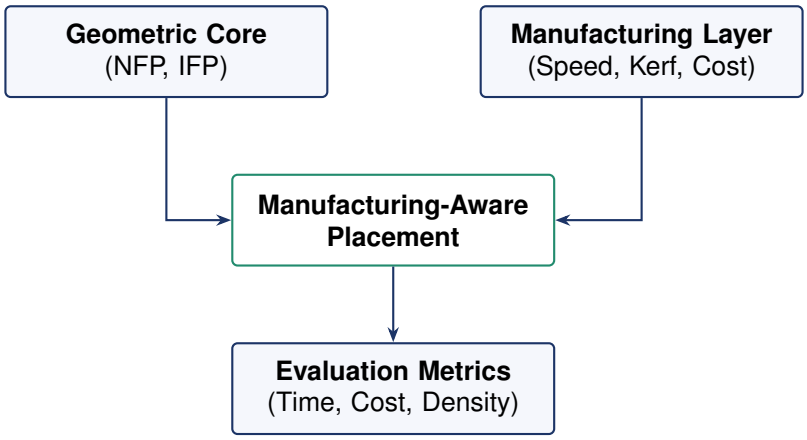
3. Objectives of the Work

The objectives of this project are as follows:

- To develop a robust 2D nesting system using computational geometry techniques
- To incorporate manufacturing constraints such as machine parameters and material properties
- To modify placement decisions by considering cutting effort and operational cost
- To evaluate nesting layouts using both geometric and manufacturing metrics
- To compare geometric and manufacturing-aware nesting strategies

4. System Overview

The system is designed as a modular architecture consisting of a geometric nesting core and a manufacturing-aware evaluation layer.



4.1 Geometric Nesting Core

The geometric engine is responsible for generating feasible layouts using:

- No-Fit Polygon (NFP): ensures that parts do not overlap
- Inner Fit Polygon (IFP): ensures that parts remain within the sheet boundary

A greedy bottom-left placement strategy is used to position parts sequentially, with support for multiple orientations.

This layer ensures:

- Collision-free placement
- Valid positioning within the sheet
- Efficient packing of shapes

4.2 Manufacturing Layer

A manufacturing layer is introduced to model real production conditions. This includes:

- Machine parameters such as:
  - Cutting speed
  - Rapid travel speed
  - Kerf width
  - Piercing cost
- Material properties such as:
  - Sheet dimensions
  - Cost per sheet

This layer enables evaluation of layouts in terms of manufacturing performance.

4.3 Manufacturability Constraints

Before placement, all shapes are validated to ensure they are physically manufacturable. The following checks are applied:

- Removal of degenerate or zero-area shapes
- Enforcement of minimum feature size based on machine type
- Verification that shapes can fit within the sheet dimensions

This ensures that all generated layouts are feasible for actual production.

4.4 Manufacturing-Aware Placement Strategy

The placement process is extended by incorporating manufacturing considerations into the scoring function.

Each candidate placement is evaluated based on:

- Geometric compactness
- Cutting effort (estimated using perimeter)
- Travel effort (distance between placements)

This allows the system to make placement decisions that consider both spatial efficiency and manufacturing cost.

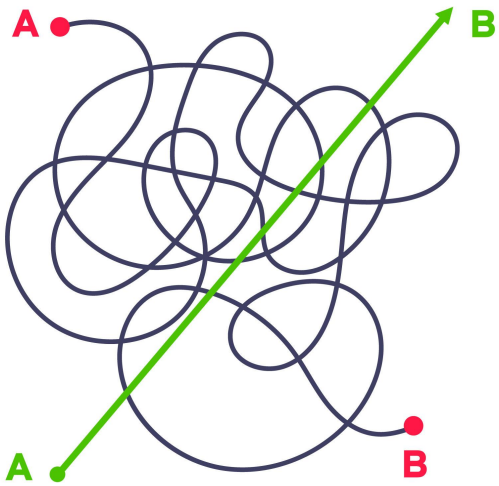


Fig: Comparison of inefficient and optimized toolpaths showing reduced travel distance

As illustrated in the figure, an inefficient cutting sequence may involve long travel distances between parts, increasing non-productive machine movement. In contrast, an optimized sequence clusters operations spatially, reducing rapid travel distance. This directly contributes to lower machining time and improved overall efficiency, which is incorporated into the placement scoring mechanism in this work.

4.5 Post-Processing and Evaluation

After generating a layout, the system computes the following metrics:

- **Material Metrics**
  - Fixed-sheet utilisation

    - Packing density
    - Scrap area
- **Cutting Metrics**
    - Total cutting length
    - Rapid travel distance
    - Number of piercings
- **Time Metrics**
    - Cutting time
    - Rapid movement time
    - Total machine time
- **Cost Metrics**
    - Material cost
    - Machine operation cost
    - Total production cost
- **Sequencing**
    - Approximate cutting sequence based on tool movement

4.6 Comparative Analysis

The system evaluates two approaches:

- Geometric Nesting (baseline)
- Manufacturing-Aware Nesting (proposed)

The comparison highlights trade-offs between packing efficiency and manufacturing performance.

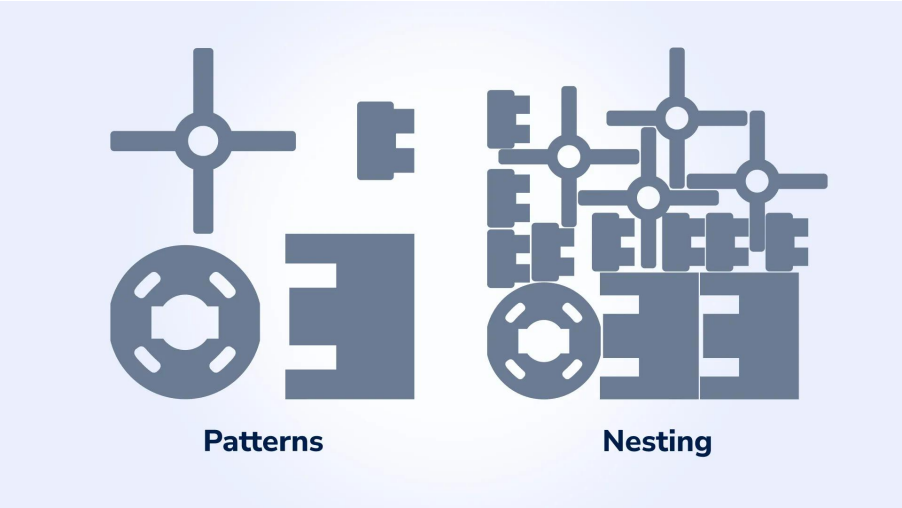


Figure 1: *
Fig: Example of nested layout for sheet cutting optimisation

5. Key Contributions

The main contributions of this work are:

- Integration of manufacturing-aware scoring into the placement process
- Development of a modular system without modifying the core geometric engine
- Introduction of dual utilisation metrics (packing density and fixed-sheet utilisation)
- Implementation of cost-based evaluation of nesting layouts
- Comparative analysis of geometric and manufacturing-aware approaches

6. Intended Outcomes

The expected outcomes of this project include:

- A functional nesting system capable of handling arbitrary polygonal shapes
- A framework for evaluating layouts using both geometric and manufacturing metrics
- Demonstration that maximum packing does not always minimise manufacturing cost
- Insight into the trade-offs between material utilisation and production efficiency

7. Assumptions

The current system is based on the following assumptions:

- Cutting speed is constant for a given machine
- Kerf width is uniform across the sheet
- Thermal effects and material deformation are neglected
- The problem is treated as a 2D planar cutting process

8. Limitations

The system has the following limitations:

- Greedy placement strategy may not achieve global optimality
- Toolpath modelling is approximate and does not include advanced cutting strategies
- Real-time constraints such as clamping and fixture limitations are not considered
- Limited differentiation between solutions for small or low-density layouts

9. Scope for Future Work

Future improvements can include:

- Integration with CAD/CAM pipelines (DXF input, G-code generation)
- Advanced toolpath planning with lead-in and lead-out strategies
- Inclusion of thermal and material deformation effects
- Multi-sheet nesting and batch optimisation
- Hybrid optimisation methods combining heuristics and metaheuristics

10. Conclusion

This project extends traditional nesting by incorporating manufacturing considerations into the placement process. By combining geometric optimisation with cost and process awareness, the system provides a more realistic evaluation of nesting performance.

The work demonstrates that efficient manufacturing requires balancing material utilisation with cutting effort and operational cost.

11. References

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